

Twelve systems, two questions each

Part A, in session 2, on paper and in pairs: classify twelve systems with the two-question test. Part B, after session 3: run one prompt on two frontier models and report where they diverge.

Worksheet

Pairs

No machine in Part A

Part B due session 4

Before you start

A · twelve systems

Part A · the nested boxes

B · one prompt, two...

Reflection

Before you start

Part A is done on the printed sheet with no laptop open. The notebook version exists only so the instructor's classification can be shown after the discussion, and so Part B can be recorded in one place.



By the end of this session you can

- Apply the two-question test to an unfamiliar system.
- Say whether a system's competence comes from people, from search, or from data.
- Explain why AI, machine learning, deep learning and LLMs are nested, not synonyms.
- Compare two frontier models on one prompt without confusing fluency with evidence.

PART A · SESSION 2

twelve systems

A

For each system answer the two questions from session 1.

Q1. Does the task require behaviour we would normally call intelligent? **Q2.** If yes, where did the operational rules come from — **people** (rules), **search**, or **data** (learning)?

#	System	Q1: intelligent?	Q2: people · search · data	One-line reason
1	A tax form checker that applies the published rules			
2	A chess engine that explores millions of moves			
3	A spam filter trained on labelled email			
4	A GPS route planner over a road map			
5	A 1970s medical expert system (MYCIN)			
6	Face unlock on a phone			
7	A thermostat			
8	A pocket calculator			
9	A chatbot such as ChatGPT or Claude			
10	A Mobile Money fraud alert			
11	A Sudoku solver			
12	Automatic translation between French and Dioula			



Two rows have no single right answer. Find them, and write down what extra fact about the system would settle it.

SESSION 2-3

Part A · the nested boxes

Complete each statement in your own words.

1. A rule-based expert system can be AI without being ____.
2. A neural network is one kind of ____.

3. A large language model is a **_ model whose rules came from _**.

4. "AI" is the **broadest / narrowest** term of the four. Circle one and explain.

after the discussion

PYTHON

```
# shown after the discussion – the instructor's reading, not the only defensible one
reading = {
    1: ("yes", "people"), 2: ("yes", "search"), 3: ("yes", "data"), 4: ("yes", "search"),
    5: ("yes", "people"), 6: ("yes", "data"), 7: ("no", "-"), 8: ("no", "-"),
    9: ("yes", "data"), 10: ("yes", "data – or people, if it is a fixed set of rules"),
    11: ("debatable", "search"), 12: ("yes", "data"),
}
for k, (q1, q2) in reading.items():
    print(f"{k:>2}  Q1 {q1:<10} Q2 {q2}")
```

what it should print

OUTPUT

```
1  Q1 yes      Q2 people
2  Q1 yes      Q2 search
3  Q1 yes      Q2 data
4  Q1 yes      Q2 search
5  Q1 yes      Q2 people
6  Q1 yes      Q2 data
7  Q1 no       Q2 –
8  Q1 no       Q2 –
9  Q1 yes      Q2 data
10 Q1 yes      Q2 data – or people, if it is a fixed set of rules
11 Q1 debatable Q2 search
12 Q1 yes      Q2 data
```

PART B · AFTER SESSION 3

one prompt, two frontier models

B

Choose **one** question you can check yourself — about Côte d'Ivoire, your field, or this course. Send exactly the same prompt to two different frontier models. Do **not** paste their answers here; summarise them.

The exact prompt
Models used, with version if shown
One important agreement
One important divergence
Which answer you trust more, and why
One fact you checked independently, and what you found

Reflection



Write it in the notebook, in your own words

In 120–180 words: why is “*does it use AI?*” often a less useful question than “*how does it produce its decisions?*”

Your answer: